

MARITIME COMPLIANCE TOOLS

USER GUIDE

EEXI Maritime Calculator—Web Version



EEXI Maritime Calculator - Attained and Required EEXI

Prepared for: Vessel Compliance & Reporting Teams

Document version: 1.0

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1. Introduction

1.1 Purpose of this guide

This guide explains how to use the EEXI Maritime Calculator, a browser-based tool for calculating a vessel's Energy Efficiency Existing Ship Index (EEXI) — the Attained EEXI and the Required EEXI — in line with IMO Resolution MEPC.350(78) (attained EEXI formula and reference-speed determination) and MEPC.364(79) (Required EEXI reference lines and correction factors). It walks through every section of the tool, the required inputs, the correction factors, and the compliance results, using screenshots taken directly from the calculator.

The guide is written for ship operators, technical superintendents, and compliance officers who need to enter ship, engine and reference-speed data, apply the relevant correction factors, run an EEXI calculation, and produce a shareable compliance report.

1.2 What the tool does

- Calculates the vessel's Attained EEXI from main-engine and auxiliary-engine particulars, fuel type, SFC and reference speed, per Resolution MEPC.350(78).
- Calculates the Required EEXI reference line for the vessel's ship type and Deadweight, per MEPC.364(79), with the correct phase-in reduction factor.
- Supports four Vref (reference speed) determination cases: EEDI ship, estimated speed-power curve, tank test + sea trial, and the statistical approximation method — including a built-in Speed-Power Curve Digitizer.
- Applies the full set of MEPC.364(79) correction factors (fc, fl, fi, fm, fj) with dedicated calculators for each ship-type-specific sub-case.
- Plots an interactive Required EEXI line against the vessel's Attained EEXI, and reports a compliance margin and status.
- Exports/imports full case data as JSON, and produces a print-ready PDF/Word compliance report.

Before you start

- The tool runs entirely in your browser — no data is uploaded or sent anywhere.
- Have the vessel's engine test/shop-trial data, NOx Technical File, Bunker Delivery Notes, and any sea-trial or model-test speed-power data ready.
- Use a current version of Chrome, Edge, or Firefox for the best experience.

2.Interface Overview

The calculator is organised into a fixed top toolbar and four numbered input sections — A: Ship Particulars, B: Main Engine & Auxiliary Engine, C: Reference Speed Vref, and D: Correction Factors — followed by a Results & Compliance Outlook section that appears once you click Calculate.



Figure 1 — Top toolbar: Calculate, Export JSON, Import, Print / PDF, Reset and Help.

The toolbar buttons are available at all times and are described below.

Button	Function
► Calculate	Runs every EEXI calculation using the current Section A-D inputs and populates the Results & Compliance Outlook section (worksheets, chart, KPI cards, compliance indicator, results table). Re-click after any change to refresh the results.
Export JSON	Downloads all current input values (ship particulars, engines, Vref case data, correction factors) as a structured JSON file — use it to archive a session or share it with a colleague.
Import	Opens a file picker to load a previously exported JSON file. Every field — including fuel rows, Vref case selection and correction-factor settings — is restored exactly as it was when exported.
Print / PDF	Generates a formatted A4 compliance report containing all inputs and the attained/required EEXI results. Use your browser's Print → Save as PDF. Run Calculate first so the report includes results.
Reset	Clears every input field back to blank/default and hides the Results section. Use this to start a completely fresh calculation.
? (Help)	Opens the built-in page guide — a complete on-screen reference of every workflow step, toolbar button, and input/output field.

Note The two circular icons at the top-left of the page (back arrow and home) are simple navigation shortcuts and do not affect your entered data.

3.Quick Start Workflow

Follow these steps for a typical single-vessel EEXI calculation:

Step	Action	What to do
1	Enter Ship Particulars	In Section A, fill in General Information (shipowner, shipbuilder, hull no., IMO no., Ship Type, Ice class) and Principal Particulars (LOA, LBP, breadth, depth, summer draught, Deadweight).
2	Enter Engine Data	In Section B, enter MCRME, RPM, SFC and fuel type for the main engine(s), and the same for the auxiliary engine(s). Configure power limitation or a shaft generator if fitted.
3	Determine Vref	In Section C, choose the applicable Vref determination case and use the pencil icon to enter or confirm the reference speed for that case.
4	Configure Correction Factors	In Section D, tick Apply on any correction-factor card that applies to your vessel. Any factor left un-applied defaults to 1.0000 (no correction).
5	Calculate & Review	Click ► Calculate in the toolbar. Review the Attained EEXI, Required EEXI, compliance chart and KPI cards in the Results section.
6	Export & Report	Use Export JSON to archive the case, and Print / Save PDF to generate the formal compliance report for distribution.

4. Section A — Ship Particulars

Section A captures the vessel's identity and principal particulars. Free-text identity fields appear on the printed report only; Ship Type, Ice class and Deadweight actively drive the calculation throughout the rest of the tool.

Ship Particulars
GENERAL INFORMATION AND PRINCIPAL PARTICULARS

General Information
Ownership and vessel identity details

SHIPOWNER: Enter shipowner
SHIPBUILDER: Enter shipbuilder
HULL NO.: Enter hull number
IMO NO.: Enter IMO number
SHIP TYPE: — Select ship type —
ICE CLASS: N/A

Principal Particulars
Main dimensions and deadweight

LENGTH OVERALL (M): e.g. 229.00
LENGTH BP (M): e.g. 222.00
BREADTH (M): e.g. 32.26
DEPTH (M): e.g. 19.60
SUMMER DRAUGHT (M): e.g. 14.50
DEADWEIGHT (T): e.g. 82000

Figure 2 — Section A populated with sample vessel particulars.

Field	Description
Shipowner / Shipbuilder / Hull No. / IMO No.	Free-text vessel identity details. Appear on the printed PDF report only — they do not affect any calculation.
Ship Type	Selects the vessel's MEPC.364(79) ship-type category. Determines the reference-line constants (a, exponent c), the reduction-factor table, and which specialty correction factors in Section D are potentially applicable (e.g. gas carrier, bulk carrier, ro-ro).
Ice class	N/A, IA Super, IA, IB or IC. Feeds the fm (ice-classed ships) factor and the ice-classed branch of fj in Section D.
Length overall / Length BP / Breadth / Depth / Summer draught (m)	Principal hull dimensions. Stored on the report; certain specialty correction-factor formulas (e.g. ro-ro fjRoRo, cranes / side-loader / ro-ro ramp cards) draw on breadth and draught.
Deadweight (t)	The vessel's summer deadweight — the primary capacity value used as the denominator throughout the attained EEXI calculation (transport-work term), in the Required EEXI reference-line formula, and as the DWT input to every cubic-capacity / capacity-based correction factor.

5. Section B — Main Engine & Auxiliary Engine

Section B records the engine particulars that feed the attained EEXI numerator (CO₂ emissions terms), split into a Main Engine (ME) card and an Auxiliary Engine (AE) card.

B Main Engine & Auxiliary Engine
ENGINE PARTICULARS FOR ATTAINED EEXI INPUTS

ME Main Engine
Primary propulsion engine details

MANUFACTURER: MAN Energy Solutions TYPE: 6G60ME-C9.5 NUMBER OF SETS: 1

OVERRIDABLE POWER LIMITATION: No MCR_{ME} (kW): 10800 RPM AT MCR: 90

SFC INPUT METHOD: Manual entry SFC AT 75% OF MCR_{ME}: 171.0 FUEL TYPE: HFO C_F: 3.114

SHAFT GENERATOR: No shaft generator

P_{ME} IN KW: 8,100.000
PME = 75% × MCR_{ME}

AE Auxiliary Engine
Electrical generation / auxiliary engine details

MANUFACTURER: Yanmar TYPE: 6EY26W NUMBER OF SETS: 3

MCR_{AE} (kW): 900 RPM AT MCR: 900

SFC INPUT METHOD: Manual entry SFC AT 50% MCR_{AE}: 215.0 FUEL TYPE: HFO C_F: 3.114

P_{AE} IN KW: 520.000
P_{AE} per MEPC.364(79) 2.2.5.6: 2.5% × IMCRME() + 250 kW when total propulsion power is ≥ 10,000 kW, otherwise 5% × IMCRME().

Figure 3 — Section B with sample main- and auxiliary-engine particulars entered.

Field	Description
Manufacturer / Type / Number of sets	Descriptive engine identity and how many identical units are installed. Number of sets multiplies MCR when computing total installed power.
Overridable power limitation	Yes/No — whether an Engine Power Limitation (EPL) or Shaft Power Limitation (SPL) is fitted. Selecting Yes reveals the limitation type and MCR _{ME,lim} , and switches the PME basis from 0.75×MCR to 0.83×MCR _{ME,lim} .
MCR _{ME} (kW) / RPM at MCR	Rated Maximum Continuous Rating of the main engine and its rated speed. MCR _{ME} is the base value used for the PME power basis (0.75×MCR by default).
SFC input method	How the Specific Fuel Consumption value is sourced: Manual entry, calculated from NO _x Technical File values, or a fixed SFC _{app} value.
SFC at 75% MCR _{ME} / 50% MCR _{AE}	Specific fuel consumption (g/kWh) at the relevant load point, used together with CF to compute the CO ₂ emission terms.
Fuel type / CF	Select the fuel consumed; CF (tonnes CO ₂ per tonne fuel) auto-populates from the MEPC.364(79) carbon-factor table for that fuel and is read-only.

Field	Description
Shaft generator / MCR PTO unit (kW)	If a Power Take-Off (PTO) shaft generator is fitted, select it and enter its MCR so the tool deducts the appropriate share from PME per MEPC.364(79) 2.2.5.2.
PME in kW (summary card)	Auto-calculated main-engine power basis: $MCRME \times \text{number of sets} \times 0.75$ (or $\times 0.83$ if power-limited), less any PTO deduction. Feeds CO ₂ Term 1 in the Results section.
PAE in kW (Auto / Manual toggle)	Auto calculates auxiliary-engine power per MEPC.364(79) 2.2.5.6 from total installed main-engine MCR. Manual lets you type a fixed PAE value instead.

6.Section C — Reference Speed V_{ref}

Section C determinesthevessel's reference speed,usedin the transport-work (denominator) term of the attained EEXI. Four determination cases are available, each mapped to a sub-clause of section 2.2.3 of Resolution MEPC.350(78).

Figure 4 — Section C: V_{ref} determination case selector and confirmed reference speed.

Field	Description
Vref determination case	Choose Case 1 (EEDI ship), Case 2 (estimated speed-power curve), Case 3 (tank test + sea trial, EEDI/design load draught) or Case 4 (statistical approximation method). This selects which input block becomes active.
Pencil / source-selection icon	Opens a dedicated modal for the selected case, offering three source options: manual entry of a known/verified V_{ref} value, the Speed-Power Curve Digitizer (upload a curve image, calibrate axes, trace the curve, and auto-interpolate V_{ref} at PME), or an available speed-power value table (manual entry / Excel-CSV import with interpolation).
Reference speed V_{ref} (display)	Read-only. The confirmed reference speed (knots) resulting from the selected case and source — the value actually used in the attained EEXI denominator.
Regulation reference	Records which sub-clause of section 2.2.3 of Res.MEPC.350(78) (2.2.3.1–2.2.3.7) supports the chosen V_{ref} determination method, for the printed report.
Case 3 fields — $V_{S,EEDI}$, $P_{S,EEDI}$, P_{ME} used	Sea-trial service speed and the power recorded at that speed, plus the PME from Section B. $V_{ref} = V_{S,EEDI} \times (P_{ME} / P_{S,EEDI})^{1/3}$.
Case 4 fields — $V_{S,service}$, $DW_{TS,service}$, $P_{S,service}$, k coefficient	Speed, deadweight and power at the design load draught / service condition. The k coefficient (auto, from ship type/size) is used together with these values in the Case 4 V_{ref} formula.

Important Whichever source you use, always click Confirm V_{ref} Source (or Apply in the Appendix modal) inside that case's modal — this is what writes the final value into Reference speed V_{ref} and refreshes the calculation.

7. Section D — Correction Factors

Section D contains five ship-condition correction factors plus a fixed weather factor. Each specialty card has its own Apply control — leave a factor un-applied and it defaults to 1.0000 (no correction).

Figure 5 — Section D: cubic-capacity, cargo-gear and capacity-limitation correction factor cards (top of section shown).

Field	Description
f_c — Cubic capacity correction	Applies to chemical tankers (§2.2.12.1), gas carriers LNG/LPG as f _{cLNG} (§2.2.12.2), ro-ro passenger ships as f _{cRoPax} (§2.2.12.3), light-cargo bulk carriers (§2.2.12.4), and ro-ro vehicle carriers as f _{cVEHICLE} (§2.2.7). Each card takes DWT/Capacity (auto from Section A) and cargo tank capacity (m³) or GT, and computes the ratio R and resulting f _c .
f₁ — Cargo-gear factor	For general cargo ships fitted with cranes, side loaders and/or ro-ro ramps. Selecting Yes for cranes / side loaders / ro-ro opens a details popup to enter their lifting/loading capacities, which feeds the combined f ₁ result.
f_i — Technical/regulatory capacity limitation	Covers ice-classed ships as f _{i,ice} (§2.2.11.1), voluntary structural enhancement as f _{iVSE} (§2.2.11.2), and CSR bulk carriers/oil tankers as f _{iCSR} (§2.2.11.3).

Field	Description
fm — Ice-classed ship factor	Applies to ships with IA Super or IA ice class, per the relevant MEPC.364(79) provision.
fj — Ship-specific design elements	Four sub-types: ice-classed ships (§2.2.8.1), shuttle tankers (§2.2.8.2, fixed fj = 0.77 for 80,000–160,000 DWT with confirmed propulsion redundancy), ro-ro ships as fjRoRo (§2.2.8.3 & §2.2.6 of MEPC.350(78)), and general cargo ships (§2.2.8.4).
fw — Weather (sea condition) correction factor	Fixed at 1.000 in this tool version (read-only) and shown for completeness in the attained EEXI worksheet.

8. Section E — Results & Compliance Outlook

This section is populated only after clicking **Calculate**. It combines a detailed calculation worksheet, an interactive chart, and headline compliance indicators, following the layout below.

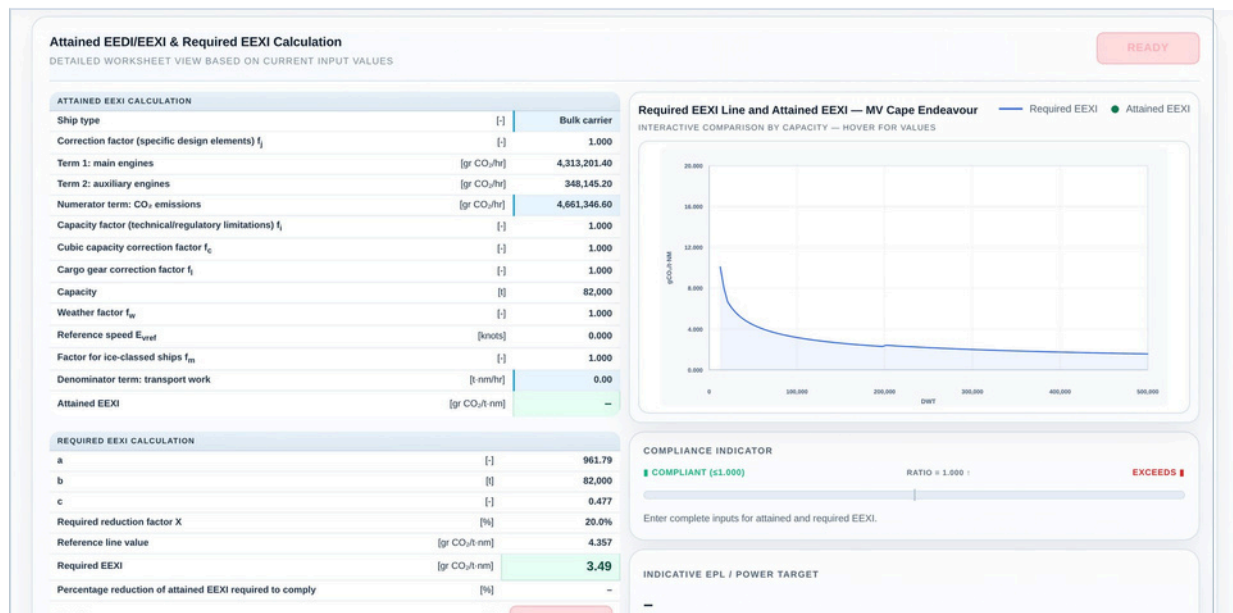


Figure 6a — Attained / Required EEXI worksheet and the Required EEXI vs. Attained EEXI chart.

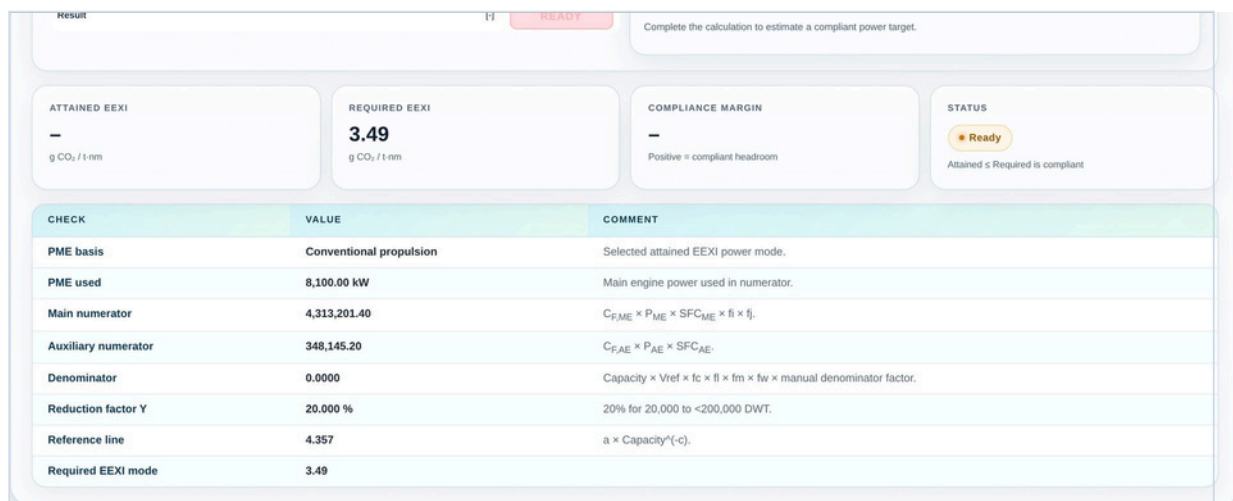


Figure 6b — Compliance Indicator, KPI cards and the results audit table.

Field	Description
Attained EEXI calculation worksheet	Row-by-row build-up: Ship type, f_j , Term 1 (main-engine CO ₂ , gr/hr), Term 2 (auxiliary-engine CO ₂ , gr/hr), Numerator (total CO ₂ emissions), f_i , f_c , f_l , Capacity, f_w , reference speed V_{ref} , f_m , Denominator (transport work, t-nm/hr), and the final Attained EEXI (g CO ₂ /t-nm).
Required EEXI calculation worksheet	Shows a, b, c, the required reduction factor X (%), the reference-line value, the final Required EEXI, the percentage reduction of attained EEXI still needed to comply, and an overall Result status.

Field	Description
Required EEXI Line and Attained EEXI (chart)	Interactive chart comparing the Required EEXI reference line against the vessel's Attained EEXI point across capacity — hover any point for exact values.
Compliance Indicator	A progress bar showing where the attained/required ratio sits: green (≤ 1.000) is compliant, red above 1.000 exceeds the required limit.
Indicative EPL / Power Target	An estimate of the maximum compliant PME and the equivalent MCR (or MCRLim) target, based on the currently selected PME mode — useful for sizing an Engine Power Limitation.
KPI cards — Attained EEXI / Required EEXI / Compliance Margin / Status	Headline numbers in g CO ₂ /t·nm, plus the Compliance Margin (positive = compliant headroom, negative = non-compliant) and a Status pill (Compliant / Exceeds / Ready).
Results table (Check / Value / Comment)	A line-by-line breakdown of every check performed, its numeric value, and a plain-language comment — a quick audit trail alongside the worksheet.

9. Exporting, Printing & Record-Keeping

- Use Export JSON after completing each case, to preserve every input value for future reuse or audit.
- Use Import to reload a previously exported JSON case rather than re-entering data.
- Click ► Calculate before using Print / PDF, so the printed report includes the attained/required EEXI results.
- Use Print / PDF and choose "Save as PDF" in the browser print dialog to generate the formal compliance report.
- Keep the exported JSON file together with the generated PDF report for full traceability.
- Name saved files consistently, e.g. VesselName_IMO_EEXI.json / .pdf, for easy retrieval.

10. Best-Practice Checklist

- Confirm Ship Type and Ice class match the vessel's actual class notation before calculating — they drive the reference-line constants and correction-factor eligibility.
 - Double-check MCR, RPM and SFC values against the engine's NOx Technical File or shop-trial report.
 - Always click Confirm Vref Source inside the Section C modal after entering or importing speed-power data — the display field is not updated until you do.
- Review which correction factors in Section D genuinely apply to the vessel; leaving an inapplicable factor at 1.0000 is correct, but a genuinely applicable factor left un-applied will overstate the Attained EEXI margin.
- Re-click ► Calculate after any change to Sections A-D — results do not refresh automatically.
 - Store the exported JSON input file together with the generated PDF/Word report for full traceability.

11. Troubleshooting

Symptom	Likely cause / fix
Results section does not appear after Calculate	One or more mandatory fields are empty (Ship Type, LOA, LBP, breadth, depth, summer draught, deadweight, MCR-ME or MCR-ME,lim, MCR-AE, SFC values, Vref). A banner lists the missing fields; fill them in and click Calculate again.
Vref shows as 0.000 in the results	The selected Vref case's source modal was opened but Confirm Vref Source was not clicked. Reopen the pencil icon in Section C, enter/import the data, and confirm.
A correction factor shows 1.0000 unexpectedly	The factor's Apply control has not been ticked, or the required inputs (e.g. cargo tank capacity, GT) are blank. Un-applied factors intentionally default to no correction.
Fuel CF field is blank or greyed out	CF auto-populates once a Fuel type is selected in Section B; choose "Other / Manual" if the exact fuel is not listed, or confirm a fuel type is selected.
Print / PDF opens a blank or partial report	Click Calculate first; the tool blocks printing until results have been generated at least once. Allow the browser a few seconds to render before printing.
Imported JSON does not restore every field	Confirm the JSON file was produced by this tool's Export JSON function; files edited outside the tool or exported from a different version may omit newer fields.

12. Appendix — Built-in Page Help

The calculator includes an on-screen Page Guide (the ? button in the toolbar) with a complete function and column reference for every section, kept in sync with the tool itself. It is a convenient quick-reference while working in the tool, alongside this document.

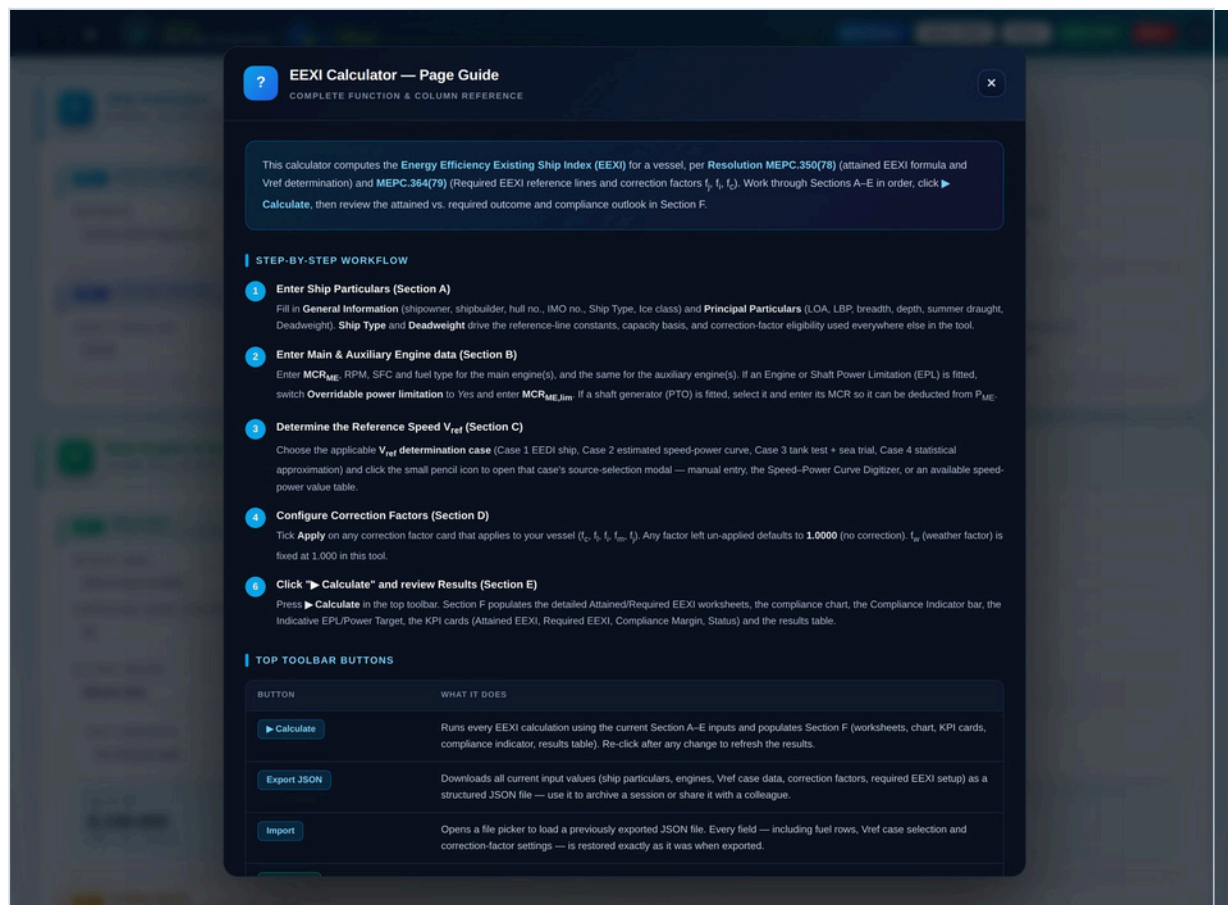


Figure 7 — The built-in Page Guide, opened from the ? button in the toolbar.

End of user guide.